V. (15 points) Magnetic fields affect electrons, so the current-voltage relation may depart from the familiar Ohm's law. This problem studies one such example.

A 2D electron gas in plane xOy has areal density n (electrons per unit area). Along positive x is uniform $\boldsymbol{E} = E\boldsymbol{i}$ （$\boldsymbol{i}$ is the x unit vector); along normal z, the magnetic field is uniform, $\boldsymbol{B} = B\boldsymbol{k}$ （$\boldsymbol{k}$ is the z unit vector). Scattering drag is proportional to velocity $\boldsymbol{v}$, modelled with time parameter τ as $\frac{-m\boldsymbol{v}}{\tau}$, m is the electron mass. Find the steady x and y current densities (current per unit transverse length) j_x and j_y using charge magnitude e、n、m、E、τ and ω, where $\omega = \frac{eB}{m}$.